

Day 4 — Assignment

Exploratory Data Analysis & Combining DataFrames — winequality-red.csv

Work through these tasks using only what we covered today. Target time: 30–45 minutes. We're back on red wine — remember it's semicolon-separated.

Section 1 — EDA Practice (mirrors 06_Pandas_practice_and_EDA)

Task 1: Read `winequality-red.csv`, then clean the column names: make every name lowercase and replace any spaces with underscores.

Task 2: Get the frequency of each quality score with `value_counts()`, then plot it as a pie chart.

Task 3: Plot that same frequency as a bar chart instead.

Task 4: Create a new column called `high_alcohol` that's True wherever a wine's alcohol is above the column's own mean. Build a crosstab of quality against `high_alcohol`, including margins (row/column subtotals).

Task 5: Investigate the `residual_sugar` column: find its minimum, maximum, and how many rows have a value above 8. Look at the 5 highest values along with their quality scores — do the sweetest wines tend to score especially high or low?

Task 6: Plot a histogram of the full `residual_sugar` column. Then plot a second histogram using only rows where `residual_sugar` is 4 or below. Which one actually shows you something useful, and why?

Section 2 — Combining DataFrames (mirrors 07_Combining_DataFrames)

Task 7: Use `pd.get_dummies()` on the quality column (prefix the new columns with 'q'), and look at the first 3 rows of the result.

Task 8: Use `.join()` to attach those dummy columns back onto the original red wine DataFrame. Confirm the new shape and column list.

Task 9: Build two small summary tables: the average citric acid by quality score, once for red wine and once for white wine (`winequality-white.csv`), using `groupby()` and `reset_index()`.

Task 10: Merge those two summary tables together on quality, using suffixes so you can tell the red and white columns apart.

Task 11: Build two small DataFrames from scratch: one with columns `product_id` and name for products p1 through p4; another with columns `product_id` and sales for

products p1 through p6 (so p5 and p6 don't exist in the first table). Merge them once with the default join, and once with how='outer'. What's different between the two results, and why?